



# Machine Learning

Sem 2-2025/2026  
Informatics Engineering Study Program  
School of Electrical Engineering and Informatics

Institute of Technology Bandung



# Course Overview

# Lecturers

- Fariska Zakhralativa Ruskanda (K1)

[fariska@informatika.org](mailto:fariska@informatika.org)

- Masayu Leylia Khodra (K2)

[masayu@staff.stei.itb.ac.id](mailto:masayu@staff.stei.itb.ac.id)

- Ayu Purwarianti (K3)

[ayu@itb.ac.id](mailto:ayu@itb.ac.id)

# Objectives For Students – IF3270

- CPMK1. Menjelaskan perbedaan jenis pembelajaran supervised learning dan reinforcement learning
- CPMK2. Mengimplementasikan algoritma backpropagation untuk Feed Forward Neural Network dan forward propagation untuk Convolutional dan Recurrent Neural Networks
- CPMK3. Menganalisis jenis pembelajaran yang tepat untuk kasus persoalan/ aplikasi tertentu
- CPMK4. Melakukan evaluasi terhadap kinerja suatu algoritma pembelajaran pada kasus persoalan tertentu

# Student Outcome - IF

Graduates of the program will have an ability to:

1. An ability to analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
2. An ability to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. An ability to communicate effectively in a variety of professional contexts.
4. An ability to recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. An ability to function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. An ability to apply computer science theory and software development fundamentals to produce computing-based solutions.

# Program Educational Objectives (PEO) - IF

1. Our graduates will have successful careers in their profession in informatics or related fields.
2. Our graduates will successfully pursue graduate study or engage in professional development.
3. Our graduates will demonstrate leadership and play active roles in the improvement of their community, especially in the development of new tools, technologies and methodologies.

# Course Description

- Credits: 3 credit points  
(3\*45 hours/semester)
- Prerequisites:
  - IF2123 Linear Algebra & Geometry
  - IF3170 Artificial Intelligence
- Grading Components:
  - Midterm & Final Test
  - Quizzes
  - Project Assignments
  - Exercises
  - Lab Works



Attending classes  
(obligatory)

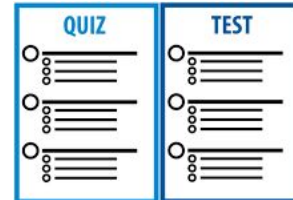
3 hours/week:

Monday 11.00-11.50;  
Thursday 09.00-10.40



Project assignments  
(groups): w4, w10

Lab works: w3,6,9,11  
Exercises



Quizzes: w5 (12 Maret  
2026), 12 (7 Mei 2026)

Midterm test: w8

Final test: w16

# General Rules



## Midterm Test & Final Test:

- obligatory, if not attending → grade E
- Additional test only for students who stay in hospital (doctor recommendation) / have “force majeure” → proof is required (from doctor, guardian)



## Any act of *cheating* will result grade “E” (for all components)

- Include helping to *cheat*
- Include cheating for assignment: “E” for all members (for all groups involved)
  - Maintain integrity between group members
- Rules for using Generative AI → explicitly written in every assignment



To pass this course, a student must have “no zero” in every component



# LMS & Communication Channel

Course Website:

<https://edunex.itb.ac.id/courses>

- Modules : IF3270 Parent Class  
Token: FB10MW  
(Token expires on 15 March 2026 23:59)
- Discussion for each class: IF3270-0x (x = class number)

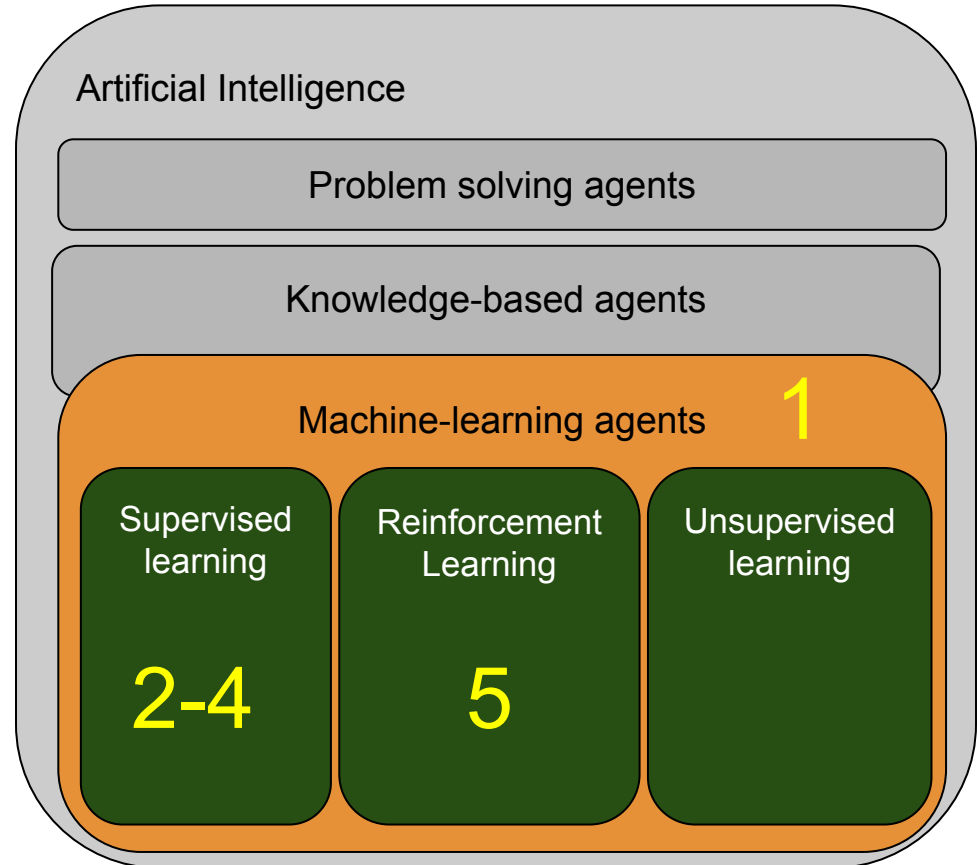
Communication:  
Join MsTeams

IF3270 Pembelajaran Mesin (2025-2)  
team code: m50gjjs

# Courses contents

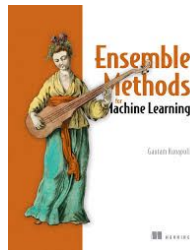


1. Machine Learning  
Overview: w1
2. Ensemble Methods: w1-w2
3. Artificial Neural Networks  
(Perceptron, FFNN, CNN,  
RNN): w3-12
4. Intro transformers: w13
5. Reinforcement learning:  
w14-15

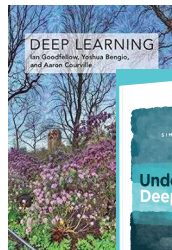


# Textbooks

1. Machine Learning Overview: w1
2. Ensemble Methods: w1-w2
3. Artificial Neural Networks (Perceptron, FFNN, CNN, RNN): w3-12
4. Intro transformers: w13
5. Reinforcement learning: w14-15



Kunapuli, G. (2023). *Ensemble methods for machine learning*. Simon and Schuster.



- Bengio, Y., Goodfellow, I., & Courville, A. (2017). *Deep learning* (Vol. 1, pp. 23-24). Cambridge, MA, USA: MIT press.
- Prince, S. J. (2023). *Understanding deep learning*. MIT press.
- Mitchell, T.,  
Machine Learning, 1997, McGraw-Hill



# Machine Learning Overview

# Jobs on The Rise 2026 (US)



*Can also be known as Machine learning engineers*

**What they do:** Artificial intelligence engineers develop and implement AI models that perform complex tasks typically requiring human decision-making, like problem-solving and prediction. | **Most common skills:** LangChain, Retrieval-Augmented Generation (RAG), PyTorch | **Most common industries:** Technology and Internet, IT Services and IT Consulting, Business Consulting and Services | **Where the most jobs are:** San Francisco, New York City, Dallas | **Current gender distribution:** 23% female, 77% male | **Median years of prior experience:** 3.7 | **Top roles transitioned from:** Software Engineer, Data Scientist, Full Stack Engineer | **Flexible work availability:** 26.2% remote, 27.1% hybrid

## Jobs on The Rise 2024 (ID)



### 6. Machine Learning Engineer

**Kegiatan pekerjaan:** Machine Learning Engineer merancang dan mengembangkan model machine learning, menerapkan algoritma kecerdasan buatan dan sistem yang dapat berjalan sendiri untuk berbagai produk dan aplikasi. | **Keahlian utama:** TensorFlow, **Machine Learning**, Deep Learning | **Industri paling umum:** Jasa TI dan Konsultan TI, Teknologi, Informasi, dan Internet, E-Learning Provider | **Lokasi lowongan kerja paling banyak:** Area DKI Jakarta, Bandung dan Sekitarnya, Malang dan Sekitarnya | **Median tahun pengalaman:** 1.8 tahun | **Paling banyak transisi dari posisi:** Data Scientist, Software Engineer, Data Engineer

# AI vs ML ?



<https://www.pastest.com/blog/medical-revision/4-facts-you-should-know-about-memory-recall/>

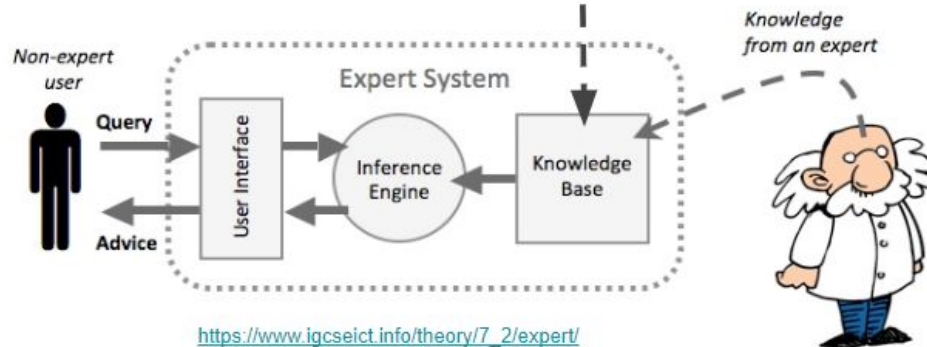
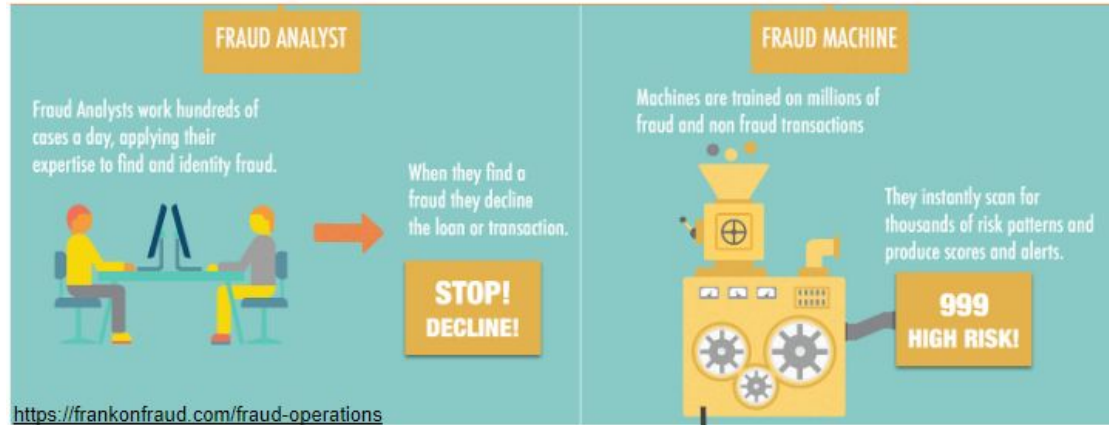
## Artificial Intelligence

Problem solving agents

Knowledge-based agents

Machine-learning agents

# ML in Knowledge-based System



An automatic knowledge acquisition technique in knowledge-based system

Knowledge is result of generalization process (inductive reasoning) in the form of data patterns

# ML Definition ?

Artificial Intelligence

Problem solving agents

Knowledge-based agents

Machine-learning agents



# ML Definition

Tom Mitchell (1998):

a computer program is said **to learn** from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , **if its performance at tasks in  $T$ , as measured by  $P$ , improves with experience  $E$ .**

Kelleher et al. (2015): an automated process that extracts patterns from data

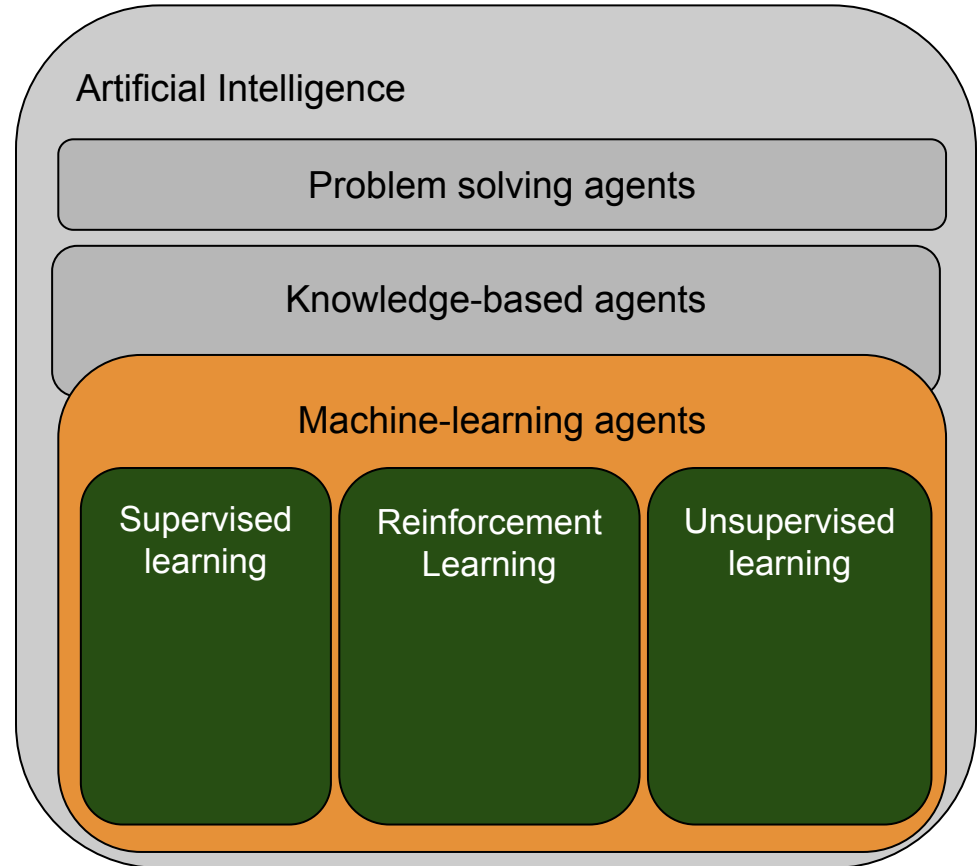
Artificial Intelligence

Problem solving agents

Knowledge-based agents

Machine-learning agents

# Learning Types



# Learning Types

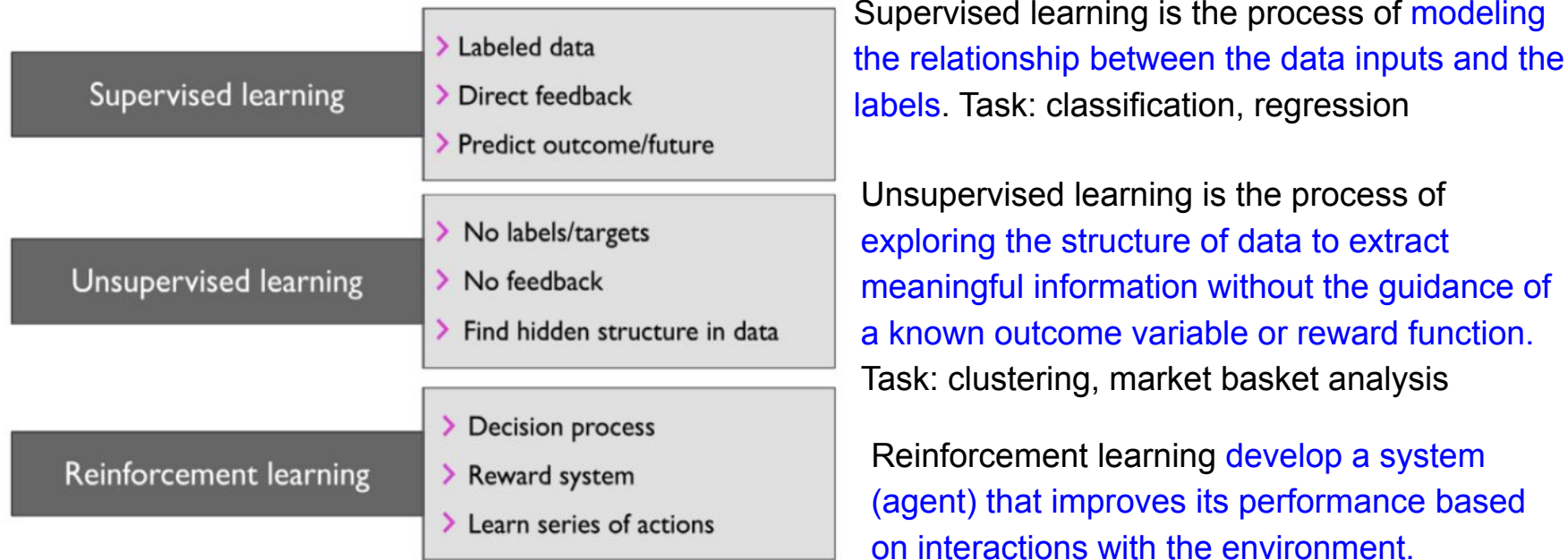


Figure 1.1: The three different types of machine learning

# Regression vs Classification

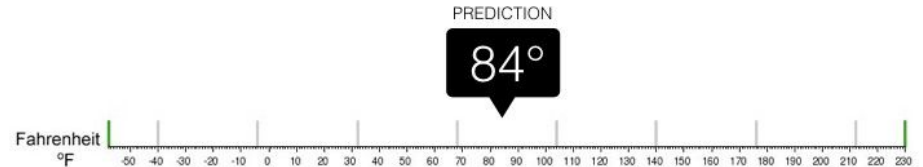
Target function  $f$ : data  $\rightarrow$  label

- Regression: domain (label) is numeric
- Classification: domain (label) is a finite set of values



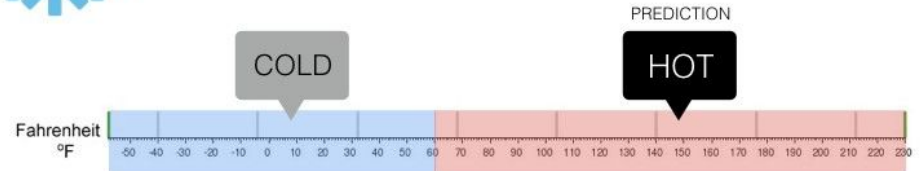
## Regression

What is the temperature going to be tomorrow?



## Classification

Will it be Cold or Hot tomorrow?



# Regression

Real world input

Model  
input

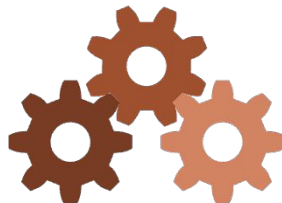
Model

Model  
output

Real world output

6000 square feet,  
4 bedrooms,  
previously sold for  
\$235K in 2005,  
1 parking spot.

$\begin{bmatrix} 6000 \\ 4 \\ 235 \\ 2005 \\ 1 \end{bmatrix}$



Supervised learning  
model

$[340]$

Predicted price  
is \$340k

# Classification

Real world input

Model  
input

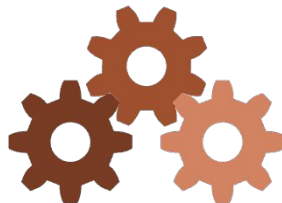
Model

Model  
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Real world output

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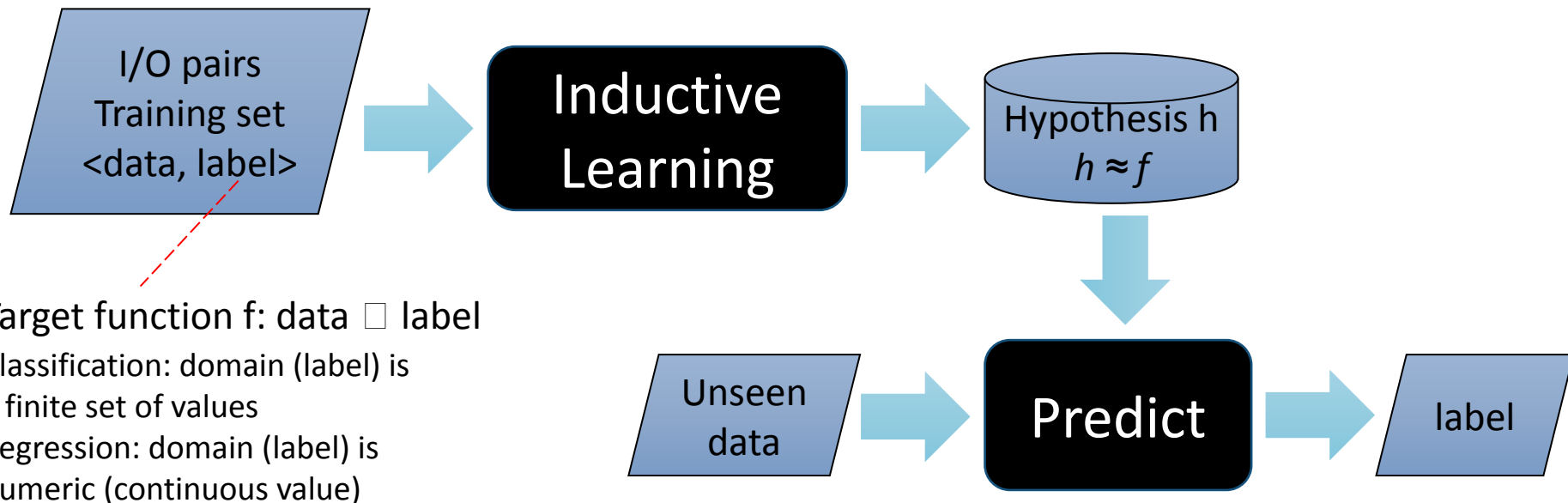
Supervised learning  
model

$[0.7, 0.3]$

Recommended: 0.7  
Not recommended: 0.3

# Supervised Learning

Learning a (possibly incorrect) general function from specific input-output pairs is called inductive learning



# Basic Terminology

- Feature
- Target/class labels
- Pattern
- Training
- Training example
- Loss function

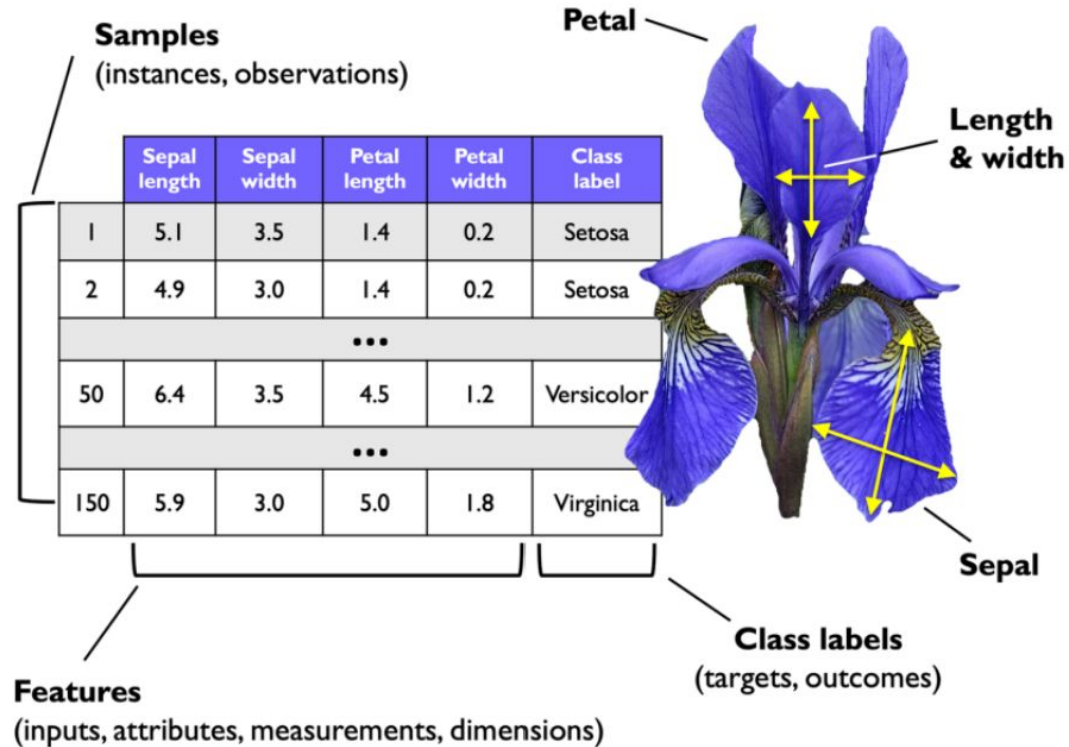


Figure 1.8: The Iris dataset



# Dataset to Pattern (Learning or Training)

| feature set |       |          |       | label |
|-------------|-------|----------|-------|-------|
| outlook     | temp. | humidity | windy | play  |
| sunny       | hot   | high     | false | no    |
| sunny       | hot   | high     | true  | no    |
| overcast    | hot   | high     | false | yes   |
| rainy       | mild  | high     | false | yes   |
| rainy       | cool  | normal   | false | yes   |
| rainy       | cool  | normal   | true  | no    |
| overcast    | cool  | normal   | true  | yes   |
| sunny       | mild  | high     | false | no    |
| sunny       | cool  | normal   | false | yes   |
| rainy       | mild  | normal   | false | yes   |
| sunny       | mild  | normal   | true  | yes   |
| overcast    | mild  | high     | true  | yes   |
| overcast    | hot   | normal   | false | yes   |
| rainy       | mild  | high     | true  | no    |



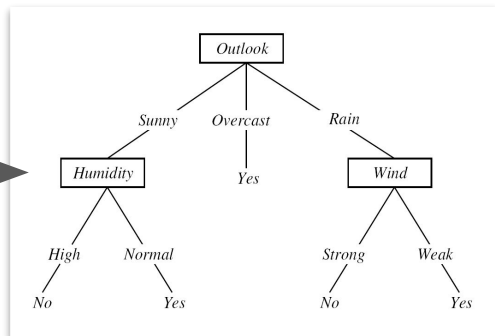
Inductive Learning

Naive Bayes

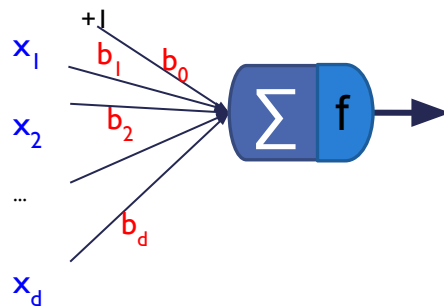
| $P(a_i   v_i)$   |              |                |               |
|------------------|--------------|----------------|---------------|
| outlook          | temperature  | humidity       | windy         |
| yes no           | yes no       | yes no         | yes no        |
| sunny 2/9 3/5    | hot 2/9 2/5  | high 3/9 4/5   | false 6/9 2/5 |
| overcast 4/9 0/5 | mild 4/9 2/5 | normal 6/9 1/5 | true 3/9 3/5  |
| rainy 3/9 2/5    | cool 3/9 1/5 |                |               |

| $P(v_i)$  |  |
|-----------|--|
| play      |  |
| yes no    |  |
| 9/14 5/14 |  |

Decision Tree Learning



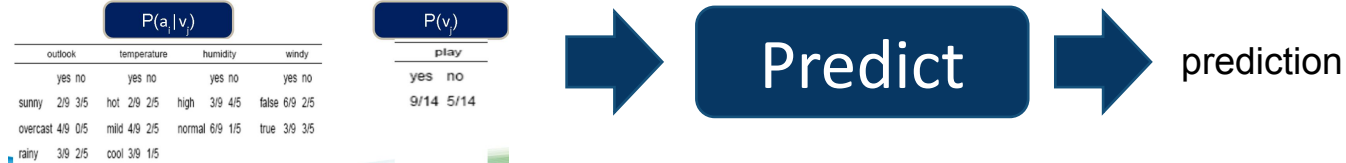
Logistic Regression



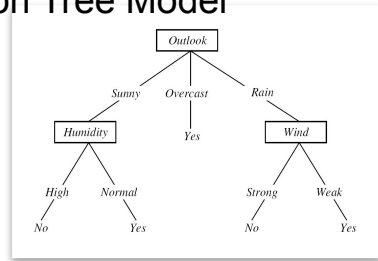
SVM

# Pattern to Decision (Inference)

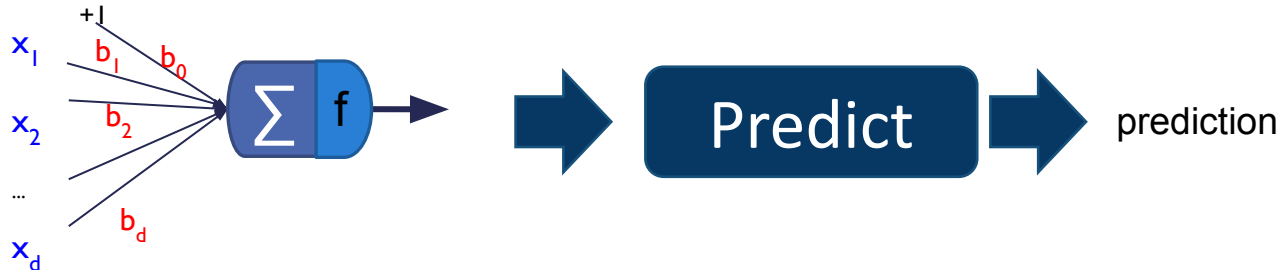
## Naive Bayes Model



## Decision Tree Model



## Logistic Regression or SVM model



# Image classification

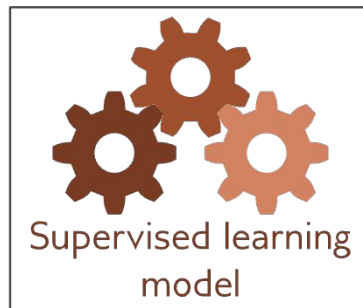
Real world input



Model  
input

$$\begin{bmatrix} 124 \\ 140 \\ 156 \\ 128 \\ 142 \\ 157 \\ \vdots \end{bmatrix}$$

Model



Model  
output

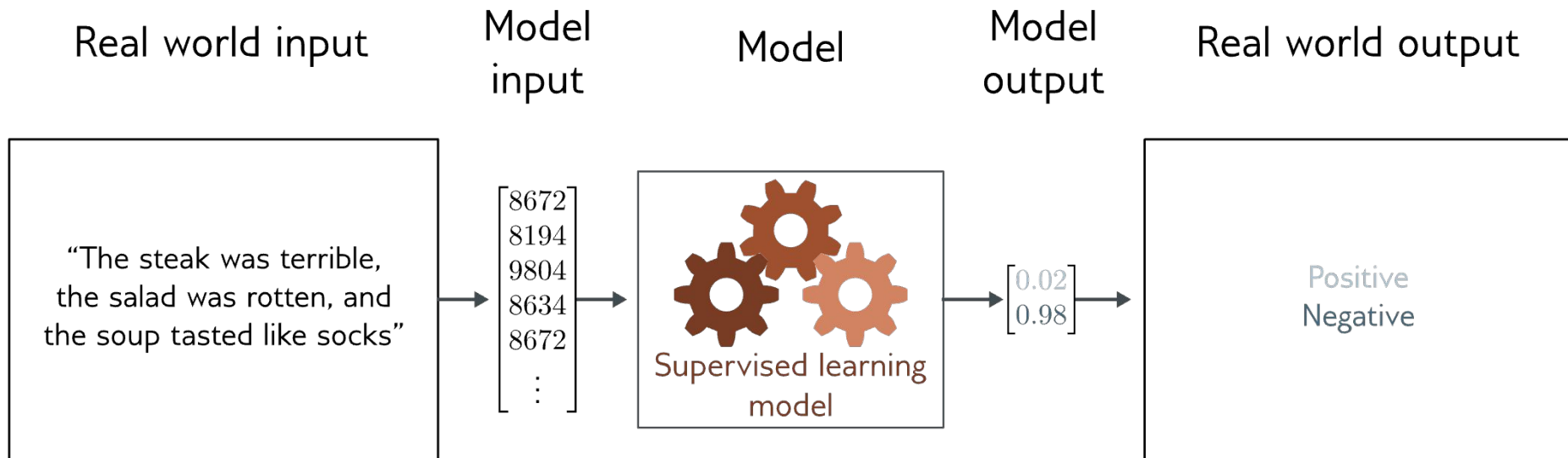
$$\begin{bmatrix} 0.00 \\ 0.00 \\ 0.01 \\ 0.89 \\ 0.05 \\ 0.00 \\ \vdots \\ 0.01 \end{bmatrix}$$

Real world output

Aardvark  
Apple  
Bee  
Bicycle  
Bridge  
Clown  
⋮

Image encoder: convolutional layer pada Convolutional Neural Network

# Text classification



Text as sequence of token, learned by Recurrent Neural Network (incl. LSTM, GRU, ReGU) or Transformer network

# Design ML System

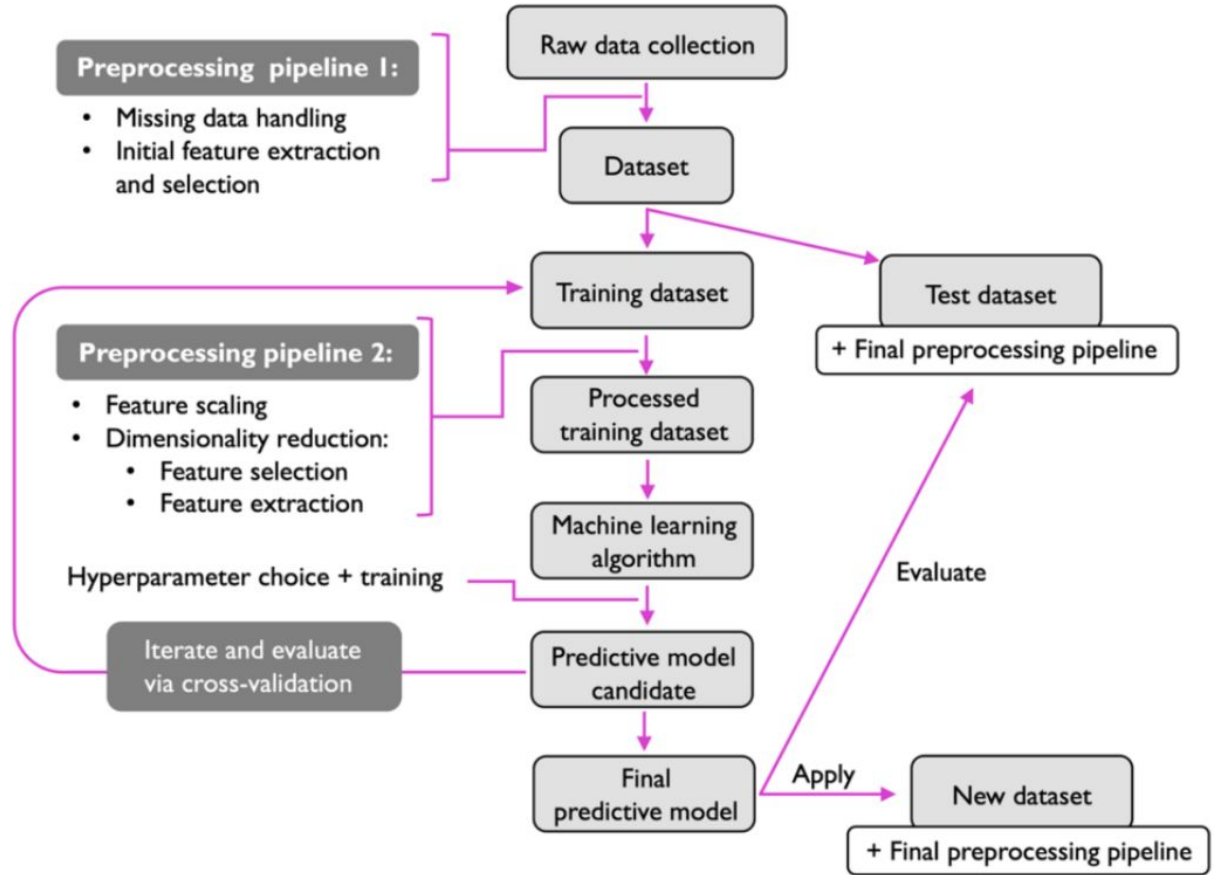


Figure 1.9: Predictive modeling workflow

Questions ?

Next: Ensemble Methods